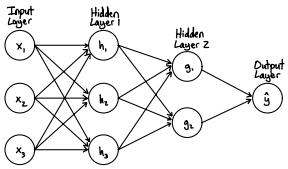
3-3-2-1 Neural Network

Objective: Neural network to predict whether a semiconductor wafer will have a defect



Define the Notation and Training Example

Input Layer:

x_1 = temperature

x_2 = production speed

x_3 = contamination level

$x_1 = 0.8$: temperature is 0.8 standard deviations above normal

$x_2 = 0.4$: production speed is 0.4 standard deviations above normal

$x_3 = 1.0$: contamination level is 1 standard deviation above normal

Hidden Layer 1:

↳ Before Activation Function: z_1, z_2, z_3

↳ After Activation Function: h_1, h_2, h_3

$h_1 = \sigma(z_1)$

$h_2 = \sigma(z_2)$

$h_3 = \sigma(z_3)$

Hidden Layer 2:

↳ Before Activation Function: z_4, z_5

↳ After Activation Function: g_1, g_2

$g_1 = \sigma(z_4)$

$g_2 = \sigma(z_5)$

Output Layer

↳ Before Activation Function: z_6

↳ After Activation Function: $\hat{y}$

$\hat{y} = \sigma(z^6)$

$\hat{y}$ = networks prediction

y = actual correct answer

↳ y { 1 Defective Wafer; 0 Good Wafer

↳ $y = 1$

Assign the Initial Weights and Biases

Input Layer to First Hidden Layer

h_1:

$w_1 = 0.10$

$w_2 = 0.20$

$w_3 = -0.10$

$z_1 = w_1 x_1 + w_2 x_2 + w_3 x_3 + b_1$

h_2:

$w_4 = -0.10$

$w_5 = 0.10$

$w_6 = 0.20$

$z_2 = w_4 x_1 + w_5 x_2 + w_6 x_3 + b_2$

h_3:

$w_7 = 0.20$

$w_8 = -0.10$

$w_9 = 0.10$

$z_3 = w_7 x_1 + w_8 x_2 + w_9 x_3 + b_3$

$$W_1 = \begin{bmatrix} 0.10 & 0.20 & -0.10 \\ -0.10 & 0.10 & 0.20 \\ 0.20 & -0.10 & 0.10 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x = \begin{bmatrix} 0.8 \\ 0.4 \\ 1.0 \end{bmatrix}$$

Entire First Layer: $z^{(1)} = W_1 x + b_1$

$$z^{(1)} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Hidden Layer 1 → Hidden Layer 2

g_1:

$w_{10} = 0.10$

$w_{11} = -0.20$

$w_{12} = 0.10$

$z_4 = w_{10} h_1 + w_{11} h_2 + w_{12} h_3 + b_4$

g_2:

$w_{13} = 0.20$

$w_{14} = 0.10$

$w_{15} = -0.10$

$z_5 = w_{13} h_1 + w_{14} h_2 + w_{15} h_3 + b_5$

$$W_2 = \begin{bmatrix} 0.10 & -0.20 & 0.10 \\ 0.20 & 0.10 & -0.10 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$h = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$$ ← First hidden layers activated outputs

Entire Second Layer: $z^{(2)} = W_2 h + b_2$

Hidden Layer 2 → Output

$w_{16} = 0.20$

$w_{17} = -0.10$

$z_6 = w_{16} g_1 + w_{17} g_2 + b_6$

$W_3 = [0.20 \quad -0.10]$

$b_3 = [0]$

$g = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}$ ← Second hidden layers activated outputs

Entire Third Layer: $z^{(3)} = W_3 g + b_3$

The Entire Network in Matrix Form:

Entire First Layer: $z^{(1)} = W_1 x + b_1$
$h = \sigma(z^{(1)})$

Entire Second Layer: $z^{(2)} = W_2 h + b_2$
$g = \sigma(z^{(2)})$

Entire Third Layer: $z^{(3)} = W_3 g + b_3$
$\hat{y} = \sigma(z^{(3)})$

Entire Forward Pass:

$x \rightarrow W_1 x + b_1 \rightarrow \sigma \rightarrow W_2 h + b_2 \rightarrow \sigma \rightarrow W_3 g + b_3 \rightarrow \sigma \rightarrow \hat{y}$

Calculate Hidden Layer 1 with Matrix Multiplication:

$z^{(1)} = W_1 x + b_1$

$$z^{(1)} = \begin{bmatrix} 0.10 & 0.20 & -0.10 \\ -0.10 & 0.10 & 0.20 \\ 0.20 & -0.10 & 0.10 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.4 \\ 1.0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$z_1 = (0.10)(0.8) + (0.20)(0.4) + (-0.10)(1.0) + 0$

$z_1 = 0.08 + 0.08 - 0.10$

$\boxed{z_1 = 0.06}$

$z_2 = (-0.10)(0.8) + (0.10)(0.4) + (0.20)(1.0) + 0$

$z_2 = -0.08 + 0.04 + 0.20$

$\boxed{z_2 = 0.16}$

$z_3 = (0.20)(0.8) + (-0.10)(0.4) + (0.10)(1.0) + 0$

$z_3 = 0.16 - 0.04 + 0.10$

$\boxed{z_3 = 0.22}$

$z^{(1)} = \begin{bmatrix} 0.06 \\ 0.16 \\ 0.22 \end{bmatrix}$ ← Entire first layer

Apply Sigmoid Function to First Layer

$h = \sigma(z^{(1)})$

$h_1 = \sigma(0.06) = \frac{1}{1 + e^{-0.06}} \approx 0.5150$

$h_2 = \sigma(0.16) = \frac{1}{1 + e^{-0.16}} \approx 0.5399$

$h_3 = \sigma(0.22) = \frac{1}{1 + e^{-0.22}} \approx 0.5548$

$h = \begin{bmatrix} 0.5150 \\ 0.5399 \\ 0.5548 \end{bmatrix}$

Calculate Hidden Layer 2 with Matrix Multiplication

$z^{(2)} = W_2 h + b_2$

$$z^{(2)} = \begin{bmatrix} 0.10 & -0.20 & 0.10 \\ 0.20 & 0.10 & -0.10 \end{bmatrix} \begin{bmatrix} 0.5150 \\ 0.5399 \\ 0.5548 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$z_4 = (0.10)(0.5150) + (-0.20)(0.5399) + (0.10)(0.5548) + 0$

$z_4 = 0.05150 - 0.10798 + 0.05548$

$\boxed{z_4 = -0.0010}$

$z_5 = (0.20)(0.5150) + (0.10)(0.5399) + (-0.10)(0.5548) + 0$

$z_5 = 0.10300 + 0.05399 - 0.05548$

$\boxed{z_5 = 0.1015}$

$z^{(2)} = \begin{bmatrix} -0.0010 \\ 0.1015 \end{bmatrix}$ ← Entire second layer

Apply Sigmoid Function to Second Layer

$g = \sigma(z^{(2)})$

$g_1 = \sigma(-0.0010) = \frac{1}{1 + e^{0.0010}} \approx 0.4997$

$g_2 = \sigma(0.1015) = \frac{1}{1 + e^{-0.1015}} \approx 0.5254$

$g = \begin{bmatrix} 0.4997 \\ 0.5254 \end{bmatrix}$

Calculate Output Layer

$z^{(3)} = W_3 g + b_3$

$z^{(3)} = [0.20 \quad -0.10] \begin{bmatrix} 0.4997 \\ 0.5254 \end{bmatrix} + [0]$

$z^{(3)} = (0.20)(0.4997) + (-0.10)(0.5254) + 0$

$z^{(3)} = 0.09994 - 0.05254$

$z^{(3)} = 0.0474$

Apply Sigmoid Function to Output Layer

$\hat{y} = \sigma(z^{(3)})$

$\hat{y} = \frac{1}{1 + e^{-0.0474}}$

$\hat{y} = 0.5119$ ← Networks initial prediction

$y = 1$ ← means the wafer is defective

So, before training, the network is only giving the defective outcome a score of 51.2%

Calculate the Binary Cross-Entropy Loss

$L = -\left[y \ln(\hat{y}) + (1 - y) \ln(1 - \hat{y})\right]$

$y = 1$
$\hat{y} = 0.5119$

$L = -\left[1 \cdot \ln(0.5119) + (1 - 1) \ln(1 - 0.5119)\right]$

$L = -\ln(0.5119)$

$L \approx 0.670$ ← If the prediction had been larger (closer to 1), the loss would have been smaller

Start Backpropagation at the Output

Goal of Backpropagation: figure out how much each weight contributed to the loss
↳ Start at the output because that is where we know the prediction error

We want to know: $\frac{\partial L}{\partial z^{(3)}}$ ← How much does the loss L change when $z^{(3)}$ changes

This is the gradient of the loss with respect to the output neuron's weighted sum $z^{(3)}$

$z^{(3)}$ affects the loss indirectly:

$z^{(3)} \rightarrow \hat{y} \rightarrow L$

Because there are two steps, use the chain rule.

$$\frac{\partial L}{\partial z^{(3)}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z^{(3)}}$$

Effect of $z^{(3)}$ on loss = effect of prediction on loss × effect of $z^{(3)}$ on prediction

Find how the prediction affects the loss:

$$\frac{\partial L}{\partial \hat{y}}$$

$$L = -\left[y \ln(\hat{y}) + (1-y)\ln(1-\hat{y})\right]$$

Substitute $y=1$

$$L = -\left[1 \ln(\hat{y}) + (1-1)\ln(1-\hat{y})\right]$$

second term goes to 0

The loss becomes:

$$L = -\ln(\hat{y})$$

Differentiate that with respect to $\hat{y}$:

$$\frac{\partial L}{\partial \hat{y}} = \frac{d}{d\hat{y}}\left[-\ln(\hat{y})\right]$$

Since:

$$\frac{d}{dx}\ln(x) = \frac{1}{x}$$

Then:

$$\frac{\partial L}{\partial \hat{y}} = -\frac{1}{\hat{y}}$$

Prediction ($\hat{y}$) was: 0.5119

$$\frac{\partial L}{\partial \hat{y}} = -\frac{1}{0.5119} \approx -1.9535$$

Since the derivative is negative, increasing the prediction $\hat{y}$ would decrease the loss.
This makes since as $y=1$ and $\hat{y} = 0.511$
This derivative is basically telling backpropagation:
The output prediction needs to move upward

Find how $z^{(3)}$ affects the Prediction

$$\frac{\partial \hat{y}}{\partial z^{(3)}}$$ ← If pre-activation value $z^{(3)}$ changes a tiny bit, how much does prediction $\hat{y}$ change?

The output neuron uses sigmoid:

$$\hat{y} = \sigma(z^{(3)})$$

The derivative of the sigmoid is:

$$\sigma'(z) = \sigma(z)(1-\sigma(z))$$

Since:

$$\hat{y} = \sigma(z^{(3)})$$

Rewrite the derivative as:

$$\frac{\partial \hat{y}}{\partial z^{(3)}} = \hat{y}(1-\hat{y})$$

Plug in prediction: $\hat{y} = 0.5119$

$$\frac{\partial \hat{y}}{\partial z^{(3)}} = (0.5119)(1-0.5119)$$

$$\frac{\partial \hat{y}}{\partial z^{(3)}} \approx 0.2499$$ ← This tells how sensitive the prediction is to the output neuron's weighted sum

If $z^{(3)}$ changes slightly, the prediction changes by roughly 0.2499 times that small change

Ex. If $\Delta z^{(3)} = 0.01$
Then $\Delta \hat{y} \approx 0.2499(0.01) \approx 0.0025$
So prediction moves from 0.5119 to 0.5144

Combine them using the Chain Rule

$$\frac{\partial L}{\partial z^{(3)}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z^{(3)}}$$

$$\frac{\partial L}{\partial z^{(3)}} = (-1.9535)(0.2499)$$

$$\frac{\partial L}{\partial z^{(3)}} = -0.4881$$ ← Around the current value of $z^{(3)}$, if $z^{(3)}$ increases slightly, the loss decreases

Ex. If $z^{(3)}$ increased by 0.01,
Then, $\Delta L = 0.01 \times -0.4881$

Output Error Term:

$\delta^{(3)} = -0.4881$
↑ The amount of error at the output neuron

Because $\delta^{(3)} < 0$

The network needs to push the output upward toward 1

This error signal $\delta^{(3)}$ is what is then sent backward to calculate how much w_{16} and w_{17} contributed to the error ← This is the starting point for the rest of backpropagation

Gradients for w_{16}, w_{17}, and Output Bias

Since we know the output error signal: $\delta^{(3)} = -0.4881$,
we can figure out how much the two output weights contributed to the loss

Output equation:

$$z^{(3)} = w_{16}g_1 + w_{17}g_2 + b_6$$

$g_1 = 0.4997$
$g_2 = 0.5254$
$w_{16} = 0.20$
$w_{17} = -0.10$

Gradient for w_{16}

$\frac{\partial L}{\partial w_{16}}$ ← Solve for this

$w_{16} \rightarrow z^{(3)} \rightarrow L$

$$\frac{\partial L}{\partial w_{16}} = \frac{\partial L}{\partial z^{(3)}} \cdot \frac{\partial z^{(3)}}{\partial w_{16}}$$

$\frac{\partial L}{\partial z^{(3)}} = -0.4881$ ← from above

Then look at:

$$z^{(3)} = w_{16}g_1 + w_{17}g_2 + b_6$$

Differentiate with respect to w_{16}, everything else is treated as constant

$\frac{\partial}{\partial w_{16}}(w_{16}g_1) = g_1$ ← The other terms $w_{17}g_2 + b_6$ do not contain w_{16}, so their derivatives with respect to w_{16} are 0

and $g_1 = 0.4997$

$$\frac{\partial}{\partial w_{16}}(0.4997\, w_{16})$$

Therefore:

$$\frac{\partial z^{(3)}}{\partial w_{16}} = g_1$$

$$\frac{\partial z^{(3)}}{\partial w_{16}} = 0.4997$$

Together:

$$\frac{\partial L}{\partial w_{16}} = (-0.4881)(0.4997)$$

$\frac{\partial L}{\partial w_{16}} = -0.2439$ ← Tells us that if w_{16} increases slightly, the loss tends to decrease (negative gradient)

Gradient for w_{17}

$\frac{\partial L}{\partial w_{17}}$ ← Solve for this

$w_{17} \rightarrow z^{(3)} \rightarrow L$

$$\frac{\partial L}{\partial w_{17}} = \frac{\partial L}{\partial z^{(3)}} \cdot \frac{\partial z^{(3)}}{\partial w_{17}}$$

$\frac{\partial L}{\partial z^{(3)}} = -0.4881$ ← from above

Then look at:

$$z^{(3)} = w_{16}g_1 + w_{17}g_2 + b_6$$

Differentiate with respect to w_{17}, everything else is treated as constant

$\frac{\partial}{\partial w_{17}}(w_{17}g_2) = g_2$ ← The other terms $w_{16}g_1 + b_6$ do not contain w_{17}, so their derivatives with respect to w_{17} are 0

and $g_2 = 0.5254$

$$\frac{\partial}{\partial w_{17}}(0.5254\, w_{17})$$

Therefore:

$$\frac{\partial z^{(3)}}{\partial w_{17}} = g_2$$

$$\frac{\partial z^{(3)}}{\partial w_{17}} = 0.5254$$

Together:

$$\frac{\partial L}{\partial w_{17}} = (-0.4881)(0.5254)$$

$\frac{\partial L}{\partial w_{17}} = -0.2565$ ← Tells us that if w_{17} increases slightly, the loss tends to decrease. (negative gradient)

Gradient for the Output Bias

$\frac{\partial L}{\partial b_6}$ ← Solve for this

$b_6 \rightarrow z^{(3)} \rightarrow L$

$$\frac{\partial L}{\partial b_6} = \frac{\partial L}{\partial z^{(3)}} \cdot \frac{\partial z^{(3)}}{\partial b_6}$$

$\frac{\partial L}{\partial z^{(3)}} = -0.4881$ ← from above

Then look at:

$z^{(3)} = w_{16} g_1 + w_{17} g_2 + b_6$

Differentiate with respect to b_6, everything else is treated as constant: $z^{(3)} = C + b_6$

$\frac{\partial z^{(3)}}{\partial b_6}(b_6) = 1$ ← Follows $\frac{d}{dx}(x) = 1$

Together:

$\frac{\partial L}{\partial b_6} = (-0.4881)(1)$

$\frac{\partial L}{\partial b_6} = -0.4881$ ← Tells us that if b_6 increases slightly, the loss tends to decrease. (negative gradient)

↳ The gradient calculation for the bias is especially simple because the bias is added directly. A tiny change of 0.01 in b_6 produces exactly a 0.01 change in $z^{(3)}$

↳ Bias changes z one-for-one

Output Layer Gradients

$\frac{\partial L}{\partial w_{16}} = -0.2439$

$\frac{\partial L}{\partial w_{17}} = -0.2565$

$\frac{\partial L}{\partial b_3} = -0.4881$

Backpropagate the Error into Hidden Layer 2 Neurons

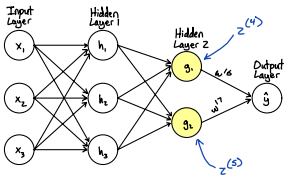
$g_1 \xrightarrow{w_6} z^{(3)}$

$g_2 \xrightarrow{w_7} z^{(3)}$

$z^{(3)} \rightarrow \hat{y} \rightarrow L$

We want to find:

$\frac{\partial L}{\partial z_4}$ and $\frac{\partial L}{\partial z_5}$



This tells us how sensitive the loss is to the weighted sums of the two neurons in Hidden Layer 2

Start with first neuron g_1

$g_1 = \sigma(z_4)$

Path from z_4 to the loss:

$z_4 \rightarrow g_1 \rightarrow z^{(3)} \rightarrow L$

$$\frac{\partial L}{\partial z_4} = \frac{\partial L}{\partial z^{(3)}} \cdot \frac{\partial z^{(3)}}{\partial g_1} \cdot \frac{\partial g_1}{\partial z_4}$$

We already know:

$\frac{\partial L}{\partial z^{(3)}} = -0.4881$

Next, the output equation was:

$z^{(3)} = w_6 g_1 + w_7 g_2 + b_6$

Therefore:

$\frac{\partial z^{(3)}}{\partial g_1} = w_{16}$

$w_{16} = 0.20$

$\frac{\partial z^{(3)}}{\partial g_1} = 0.20$

Finally, because:

$g_1 = \sigma(z_4)$

the sigmoid derivative is:

$\frac{\partial g_1}{\partial z_4} = g_1(1 - g_1)$

$g_1 = 0.4997$

$$\frac{\partial g_1}{\partial z_4} = (0.4997)(1-0.4997)$$

$$\frac{\partial g_1}{\partial z_4} = 0.2500$$

Multiply everything:

$$\frac{\partial L}{\partial z_4} = (-0.4881)(0.20)(0.2500)$$

$$\frac{\partial L}{\partial z_4} = -0.0244$$

$$\delta_4 = -0.0244$$

Second Neuron g_2

$$g_2 = \sigma(z_5)$$

Path from z_5 to the loss:

$$z_5 \rightarrow g_2 \rightarrow z^{(3)} \rightarrow L$$

$$\frac{\partial L}{\partial z_5} = \frac{\partial L}{\partial z^{(3)}} \cdot \frac{\partial z^{(3)}}{\partial g_2} \cdot \frac{\partial g_2}{\partial z_5}$$

We already know:

$$\frac{\partial L}{\partial z^{(3)}} = -0.4881$$

Next, the output equation was:

$$z^{(3)} = w_{16} g_1 + w_{17} g_2 + b_3$$

Therefore:

$$\frac{\partial z^{(3)}}{\partial g_2} = w_{17}$$

$$w_{17} = -0.10$$

$$\frac{\partial z^{(3)}}{\partial g_2} = -0.10$$

Finally, because:

$$g_2 = \sigma(z_5)$$

the sigmoid derivative is:

$$\frac{\partial g_2}{\partial z_5} = g_2(1-g_2)$$

$$g_2 = 0.5254$$

$$\frac{\partial g_1}{\partial z_4} = (0.5254)(1-0.5254)$$

$$\frac{\partial g_1}{\partial z_4} = 0.2494$$

Multiply everything:

$$\frac{\partial L}{\partial z_5} = (-0.4881)(-0.10)(0.2494)$$

$$\frac{\partial L}{\partial z_5} = 0.0122$$

$$\delta_5 = 0.0122$$

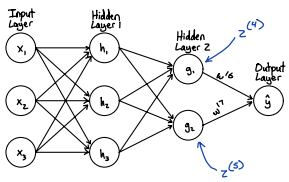
Hidden Layer 2 Error Signals

$$\delta_4 = -0.0244 = \frac{\partial L}{\partial z_4}$$

$$\delta_5 = 0.0122 = \frac{\partial L}{\partial z_5}$$

Calculate the Gradients for w_{10} through w_{15}

From the forward pass we know:



$$h_1 = 0.5150$$

$$h_2 = 0.5399$$

$$h_3 = 0.5548$$

$$z^{(4)} = w_{10} h_1 + w_{11} h_2 + w_{12} h_3 + b_4$$

Gradients for weights going into g_1

Start with w_{10}

↳ we want: $\frac{\partial L}{\partial w_{10}}$ ← How much does w_{10} affect the loss

Using Chain Rule:

$$\frac{\partial L}{\partial w_{10}} = \frac{\partial L}{\partial z_4} \cdot \frac{\partial z_4}{\partial w_{10}}$$

We already know: $\frac{\partial L}{\partial z_4} = \delta_4 = -0.0244$

Since: $z^{(4)} = w_{10}h_1 + w_{11}h_2 + w_{12}h_3 + b_4$

the derivative with respect to w_{10} is:

$\frac{\partial z_4}{\partial w_{10}} = h_1$

$h_1 = 0.5150$

$\frac{\partial L}{\partial w_{10}} = (-0.0244)(0.5150)$

$\frac{\partial L}{\partial w_{10}} = -0.0126$

The other two weights (w_{11} and w_{12}) going into z^4 work the same way:

w_{11}

$\frac{\partial L}{\partial w_{11}} = \delta_4 h_2$

$\frac{\partial L}{\partial w_{11}} = (-0.0244)(0.5399)$

$\frac{\partial L}{\partial w_{11}} = -0.0132$

w_{12}

$\frac{\partial L}{\partial w_{12}} = \delta_4 h_3$

$\frac{\partial L}{\partial w_{12}} = (-0.0244)(0.5548)$

$\frac{\partial L}{\partial w_{12}} = -0.0135$

bias

$\frac{\partial L}{\partial b_4} = \delta_4 = -0.0244$

Gradients for weights going into g_2

$z_5 = w_{13}h_1 + w_{14}h_2 + w_{15}h_3 + b_5$

Second Hidden Layer Error Signal:

$\delta_5 = 0.0122 = \frac{\partial L}{\partial z_5}$

w_{13}

$\frac{\partial L}{\partial w_{13}} = \delta_5 h_1$

$\frac{\partial L}{\partial w_{13}} = (0.0122)(0.5150)$

$\frac{\partial L}{\partial w_{13}} = 0.0063$

w_{14}

$\frac{\partial L}{\partial w_{14}} = \delta_5 h_2$

$\frac{\partial L}{\partial w_{14}} = (0.0122)(0.5399)$

$\frac{\partial L}{\partial w_{14}} = 0.0066$

w_{15}

$\frac{\partial L}{\partial w_{15}} = \delta_5 h_3$

$\frac{\partial L}{\partial w_{15}} = (0.0122)(0.5548)$

$\frac{\partial L}{\partial w_{15}} = 0.0068$

bias

$\frac{\partial L}{\partial b_5} = \delta_5 = 0.0122$

Key Pattern:

Weight gradient = neuron error signal × input entering that weight

Backpropagate the Error into Hidden Layer 1

We have the error signals for Hidden Layer 2:

$\delta_4 = -0.0244$

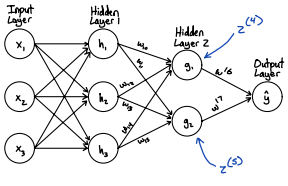
$\delta_5 = 0.0122$

Now we want the error signals for Hidden Layer 1:

$\delta_1 = \frac{\partial L}{\partial z_1}$

$\delta_2 = \frac{\partial L}{\partial z_2}$

$\delta_3 = \frac{\partial L}{\partial z_3}$



Important New Idea is that each Hidden Layer 1 Neuron feeds into both neurons in Hidden Layer 2

$h_1 \rightarrow g_1 \rightarrow L$

$h_1 \rightarrow g_2 \rightarrow L$

↳ So we have to account for both paths and add their effects together

Error Signal for h_1

↳ the two relevant weights are:

$w_{10} = 0.10$ from $h_1 \rightarrow g_1$

$w_{11} = 0.20$ from $h_1 \rightarrow g$

First, Calculate how h_1 affects the loss:

$$\frac{\partial L}{\partial h_1} = \delta_4 w_{10} + \delta_5 w_{12}$$

$$\frac{\partial L}{\partial h_1} = (-0.0244)(0.10) + (0.0122)(0.20)$$

$$\frac{\partial L}{\partial h_1} = -0.00244 + 0.00244$$

$$\frac{\partial L}{\partial h_1} \approx 0$$

Still need Sigmoid Derivative:

$h_1 = \sigma(z_1)$

$$\frac{\partial h_1}{\partial z_1} = h_1(1 - h_1)$$

$h_1 = 0.5150$

$$\frac{\partial h_1}{\partial z_1} = (0.5150)(0.4850) \approx 0.2498$$

$$\boldsymbol{\delta_1} = \frac{\partial L}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1}$$

$\boldsymbol{\delta_1} \approx 0$ ← The zero happens because the two paths from h_1 happen to almost perfectly offset one another with the weights we choose

Error Signal for h_2

↳ h_2 outgoing weights are:

$w_{11} = -0.20$

$w_{14} = 0.10$

$$\frac{\partial L}{\partial h_2} = \boldsymbol{\delta_4} w_{11} + \boldsymbol{\delta_5} w_{14}$$

$$\frac{\partial L}{\partial h_2} = (-0.0244)(-0.20) + (0.0122)(0.10)$$

$$\frac{\partial L}{\partial h_2} = 0.00610$$

Apply the sigmoid derivative:

$h_2 = 0.5399$

$h_2(1 - h_2) = (0.5399)(0.4601) \approx 0.2484$

$$\boldsymbol{\delta_2} = (0.00610)(0.2484)$$

$$\boldsymbol{\delta_2} = 0.00152$$

Error Signal for h_3

↳ h_3 outgoing weights are:

$w_{12} = 0.10$

$w_{15} = -0.10$

$$\frac{\partial L}{\partial h_3} = \boldsymbol{\delta_4} w_{12} + \boldsymbol{\delta_5} w_{15}$$

$$\frac{\partial L}{\partial h_3} = (-0.0244)(0.10) + (0.0122)(-0.10)$$

$$\frac{\partial L}{\partial h_3} = -0.00366$$

Apply the sigmoid derivative:

$h_3 = 0.5548$

$h_3(1 - h_3) = (0.5548)(0.4452) \approx 0.2470$

$$\boldsymbol{\delta_3} = (-0.00366)(0.2470)$$

$$\boldsymbol{\delta_3} = -0.000904$$

Hidden Layer 1 Error Signals

$\boldsymbol{\delta_1} \approx 0$

$\boldsymbol{\delta_2} = 0.00152$

$\boldsymbol{\delta_3} = -0.000904$

The most important concept in this step is this formula:

$$\delta_1 = (\delta_4 w_{10} + \delta_5 w_{13}) h_1 (1 - h_1)$$

The addition is because h_1 affects the loss through two different paths, so backprop adds the contributions from both.

Calculate the Gradients for w_1 through w_6

Hidden Layer 1 Error Signals

$\delta_1 \approx 0$

$\delta_2 = 0.00152$

$\delta_3 = -0.000904$

Original Inputs

$x_1 = 0.8$

$x_2 = 0.4$

$x_3 = 1.0$

Same Formula applies:

weight gradient = neuron error signal × input feeding that weight

w_1

↪ w_1 connects $x_1 \rightarrow h_1$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

Since:

$$z_1 = w_1 x_1 + w_2 x_2 + w_3 x_3 + b_1$$

$$\frac{\partial z_1}{\partial w_1} = x_1$$

Therefore:

$$\frac{\partial L}{\partial w_1} = \delta_1 x_1$$

$$\frac{\partial L}{\partial w_1} = (-0.00000157)(0.8)$$

$$\frac{\partial L}{\partial w_1} = -0.00000125$$

w_2

$$\frac{\partial L}{\partial w_2} = \delta_1 x_2$$

$$\frac{\partial L}{\partial w_2} = (-0.00000157)(0.4)$$

$$\frac{\partial L}{\partial w_2} = -0.00000063$$

w_3

$$\frac{\partial L}{\partial w_3} = \delta_1 x_3$$

$$\frac{\partial L}{\partial w_3} = (-0.00000157)(1.0)$$

$$\frac{\partial L}{\partial w_3} = -0.00000157$$

w_4

$$\frac{\partial L}{\partial w_4} = \delta_2 x_1$$

$$\frac{\partial L}{\partial w_4} = (0.001515)(0.8)$$

$$\frac{\partial L}{\partial w_4} = 0.001212$$

w_5

$$\frac{\partial L}{\partial w_5} = \delta_2 x_2$$

$$\frac{\partial L}{\partial w_5} = (0.001515)(0.4)$$

$$\frac{\partial L}{\partial w_5} = 0.000606$$

w_6

$$\frac{\partial L}{\partial w_6} = \delta_2 x_3$$

$$\frac{\partial L}{\partial w_6} = (0.001515)(1.0)$$

$$\frac{\partial L}{\partial w_6} = 0.001515$$

w_7

$$\frac{\partial L}{\partial w_7} = \delta_3 x_1$$

$$\frac{\partial L}{\partial w_7} = (-0.000904)(0.8)$$

$$\frac{\partial L}{\partial w_7} = -0.000723$$

w_8

$$\frac{\partial L}{\partial w_8} = \delta_3 x_2$$

$$\frac{\partial L}{\partial w_8} = (-0.000904)(0.4)$$

$$\frac{\partial L}{\partial w_8} = -0.000361$$

w_9

$$\frac{\partial L}{\partial w_9} = \delta_3 x_3$$

$$\frac{\partial L}{\partial w_9} = (-0.000904)(1.0)$$

$$\frac{\partial L}{\partial w_9} = -0.000904$$

Bias Gradients

$$\frac{\partial L}{\partial b_1} = \delta_1 = -0.00000157$$

$$\frac{\partial L}{\partial b_2} = \delta_2 = 0.001515$$

$$\frac{\partial L}{\partial b_3} = \delta_3 = -0.000904$$

The Gradient for every trainable parameter has now been calculated

Update the Weights and Biases using Gradient Descent

↳ Now that the gradients have been calculated, use them to change the networks parameters

Update Rule: $w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$

↳ η = learning rate

↳ The learning rate controls how large each update is

$\eta = 0.1$

Update the Output Weights

w_{16}

we had:

$w_{16} = 0.20$

and:

$$\frac{\partial L}{\partial w_{16}} = -0.2439$$

$$w_{16}^{new} = 0.20 - (0.1)(-0.2439)$$

$$w_{16}^{new} = 0.22439$$

w_{17}

$w_{17} = -0.10$

$$\frac{\partial L}{\partial w_{17}} = -0.2565$$

$$w_{17}^{new} = -0.10 - (0.1)(-0.2565)$$

$$w_{17}^{new} = -0.07435$$

Output Bias

$b_6 = 0$

$$\frac{\partial L}{\partial b_6} = -0.4881$$

$$b_6^{new} = 0 - (0.1)(-0.4881)$$

$$b_6^{new} = 0.04881$$

Update w_{10} through w_{15}

$w_{10}^{new} = 0.10126$

$w_{11}^{new} = -0.19868$

$w_{12}^{new} = 0.10135$

$w_{13}^{new} = 0.19937$

$w_{14}^{new} = 0.09934$

$w_{15}^{new} = -0.10068$

$b_4^{new} = 0.00244$

$b_5^{new} = -0.00122$

Update w_1 through w_9

$w_1^{new} = 0.10000013$

$w_2^{new} = 0.20000006$

$w_3^{new} = -0.09999984$

$w_4^{new} = -0.1001212$

$w_5^{new} = 0.0999394$

$w_6^{new} = 0.1998485$

$w_7^{new} = 0.2000723$

$w_8^{new} = -0.0999639$

$w_9^{new} = 0.1000904$

$b_1^{new} = 0.000000157$

$b_2^{new} = -0.0001515$

$b_3^{new} = 0.0000904$

Backpropagation gave $\frac{\partial L}{\partial w}$ for each weight

Gradient Descent then uses those derivates to update the weights

Backpropagation: find the gradients

Gradient Descent: use the gradients to update the parameters

Full training step:

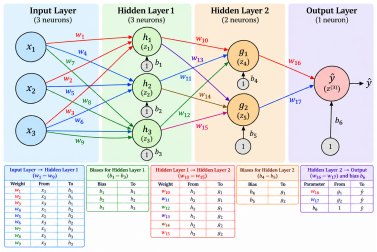
Forward Pass → Loss → Backpropagation → Gradients → Gradient Descent → New Weights

Run Another Forward Pass with the Updated Weights

↳ Test whether the network actually improved after one round of backpropagation and gradient descent

Same process as before, but using updated weights and biases:

$x \rightarrow$ Hidden Layer 1 $\rightarrow$ Hidden Layer 2 $\rightarrow \hat{y} \rightarrow L$



Input Layer → Hidden Layer 1 ($w_1 \rightarrow w_9$)		
Weight	From	To
w_1	x_1	h_1
w_4	x_2	h_1
w_7	x_3	h_1
w_2	x_1	h_2
w_5	x_2	h_2
w_8	x_3	h_2
w_3	x_1	h_3
w_6	x_2	h_3
w_9	x_3	h_3

Biases for Hidden Layer 1 ($b_1 \rightarrow b_3$)	
Bias	To
b_1	h_1
b_2	h_2
b_3	h_3

Hidden Layer 1 → Hidden Layer 2 ($w_{10} \rightarrow w_{15}$)		
Weight	From	To
w_{10}	h_1	g_1
w_{11}	h_2	g_1
w_{12}	h_3	g_1
w_{13}	h_1	g_2
w_{14}	h_2	g_2
w_{15}	h_3	g_2

Biases for Hidden Layer 2 ($b_4 \rightarrow b_5$)	
Bias	To
b_4	g_1
b_5	g_2

Hidden Layer 2 → Output ($w_{16} \rightarrow w_{17}$) and bias b_6		
Parameter	From	To
w_{16}	g_1	$\hat{y}$
w_{17}	g_2	$\hat{y}$
b_6	1	$\hat{y}$

Recalculate Hidden Layer 1 (Input → Hidden Layer 1)

Keep the same input:

$$x = \begin{bmatrix} 0.8 \\ 0.4 \\ 1.0 \end{bmatrix}$$

↳ Use updated parameters from gradient descent

$z_{new}^{(1)} = W_{1,new} x + b_{1,new}$

$$z_{new}^{(1)} = \begin{bmatrix} 0.10000013 & 0.20000006 & -0.09999984 \\ -0.1001212 & 0.0999394 & 0.1998485 \\ 0.2000723 & -0.0999639 & 0.1000904 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.4 \\ 1.0 \end{bmatrix} + \begin{bmatrix} 0.000000157 \\ -0.0001515 \\ 0.0000904 \end{bmatrix}$$

$$z_{new}^{(1)} = \begin{bmatrix} 0.0600 \\ 0.1596 \\ 0.2203 \end{bmatrix}$$

Apply Sigmoid by Element:

$h_{new} = \sigma(z_{new}^{(1)})$

$$h_{new} = \begin{bmatrix} \sigma(0.0600) \\ \sigma(0.1596) \\ \sigma(0.2203) \end{bmatrix}$$

$$h_{new} = \begin{bmatrix} 0.5150 \\ 0.5398 \\ 0.5548 \end{bmatrix}$$

Recalculate Hidden Layer 2 (Hidden Layer 1 → Hidden Layer 2)

$z_{new}^{(2)} = W_{2,new} h_{new} + b_{2,new}$

$$z_{new}^{(2)} = \begin{bmatrix} 0.10126 & -0.19868 & 0.10135 \\ 0.19937 & 0.09984 & -0.10068 \end{bmatrix} \begin{bmatrix} 0.5150 \\ 0.5398 \\ 0.5548 \end{bmatrix} + \begin{bmatrix} 0.00244 \\ -0.00122 \end{bmatrix}$$

$$z_{new}^{(2)} = \begin{bmatrix} 0.0036 \\ 0.0992 \end{bmatrix}$$

Apply Sigmoid:

$g_{new} = \sigma(z_{new}^{(2)})$

$$g_{new} = \begin{bmatrix} \sigma(0.0036) \\ \sigma(0.0992) \end{bmatrix}$$

$$g_{new} = \begin{bmatrix} 0.5009 \\ 0.5248 \end{bmatrix}$$

Hidden Layer 2 → Output

$W_{3,new} = \begin{bmatrix} 0.22439 & -0.07435 \end{bmatrix}$

$b_{3,new} = 0.04881$

$z_{new}^{(3)} = W_{3,new} g_{new} + b_{3,new}$

$$z_{new}^{(3)} = \begin{bmatrix} 0.22439 & -0.07435 \end{bmatrix} \begin{bmatrix} 0.5009 \\ 0.5248 \end{bmatrix} + 0.04881$$

$z_{new}^{(3)} = 0.1222$

Apply Sigmoid:

$\hat{y}_{new} = \sigma(z_{new}^{(3)})$

$\hat{y}_{new} = 0.5305$

Entire Forward Pass in Four Equations

$$h = \sigma\left(W_1 x + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}\right)$$

$$g = \sigma\left(W_2 h + \begin{bmatrix} b_4 \\ b_5 \end{bmatrix}\right)$$

$z^{(3)} = W_3 g + b_6$

$\hat{y} = \sigma(z^{(3)})$

Calculate the New Loss and Compare

After the second forward pass:

$\hat{y} = 0.5305$

Actual target is still:

$y = 1$

Binary Cross Entropy:

$L = -\left[y \ln(\hat{y}) + (1-y)\ln(1-\hat{y})\right]$

since $y=1$, it simplifies to:

$L = -\ln(\hat{y})$

$L_{new} = -\ln(0.5305)$

$L_{new} = 0.634$

Compare with old Loss

$L_{old} = 0.670$

↳ loss decreased

↳ The prediction also moved in the right direction:

$0.5119 \rightarrow 0.5305$

After one training iteration, prediction improved and loss decreased